

Format mirrors Pepper Square's case study template: punchy problem headline → 2 hero stats → Business Objective → Business Challenges (3 short lines) → Systems/Approach tags → What Changed (5 short items) → Transformation Scale (counts + tagline) → Business Impact (short outcome lines) → CTA to the next phase. No long paragraphs, no tables — every line is written to be scannable in seconds.

PHASE 1

Field assessors couldn't complete compliance work where the work actually happened.

[hero video/image — before/after mobile UI]

2.6× increase in field assessment completion (21% → 55%) **65%** improvement in usability score (SUS 49 → 81)

Business Objective

Give field-based compliance assessors a mobile-first way to complete inspections at the point of observation, without reverting to pen, paper, or a desktop workaround.

Business Challenges

Field assessors had no reliable way to complete assessments on-site

Manual paper workarounds caused transcription errors and compliance gaps

The existing iPad interface had a 61-point completion gap vs. desktop

Approach

UX Research · Interaction Design · Offline-First Architecture

The Research

[image: Persona — Adam, Field & Survey Officer] [image: Empathy Map] [image: As-Is Customer Journey] [image: Stakeholder Feedback Grid]

Field observation, interviews, and journey mapping surfaced the same failure point from four different angles — the moment of inspection was exactly when the app couldn't be used.

What Changed

Built offline-first assessment capture Assessors can now complete forms fully without connectivity — no data lost. *[image: To-Be Journey Map]*

Made draft recovery automatic Auto-save every 60 seconds, with persistent recovery on relaunch. *[image: high-fidelity UI — Assessment Summary screen]*

Redesigned navigation for field use Assessment launch cut from 5+ taps to 2, built for one-handed, gloved use. *[image: Balsamiq wireframes]*

Added visible sync status A persistent status bar removes the anxiety of not knowing if data saved. *[image: high-fidelity UI — Dashboard / Card view]*

Embedded photo and note capture Evidence capture now happens inside the assessment flow, not a separate app. *[image: Mobile UI 1–3]*

The Design Artifacts

[image: Feature roadmap — short-term/long-term] [image: Prioritisation grid] [image: Stakeholder map] [image: Users' hopes and fears]

Transformation Scale

6 field workflow stages redesigned end-to-end **8** field-native capabilities shipped **2** platforms unified — desktop and iPad, one consistent experience

A field-tested, offline-first assessment experience — validated, not assumed.

Business Impact

2.6× increase in field assessment completion — 21% → 55%

55% reduction in time to first submission — 62 min → 28 min

65% improvement in usability score — SUS 49 (Poor) → 81 (Excellent)

Fewer transcription errors and compliance gaps from paper workarounds

Consistent, field-native experience across desktop, iPad, and mobile

Continue to Phase 2 — Product-Led Adoption →

PHASE 2

Portfolio learning exercise — illustrative data, not collected from or delivered to the client. Builds on the real Phase 1 redesign.

A better usability score didn't prove the redesign had actually taken hold.

[hero visual — funnel diagram]

4× higher activation rate at final funnel stage (21% → 55%) **3.8×** higher 90-day retention (17% → 64%)

Business Objective

Determine whether the redesign's usability gains translated into sustained adoption over time, not just a one-time completion spike.

Business Challenges

A higher completion rate didn't confirm the feature people needed most — offline mode — was actually being adopted

No visibility into exactly where assessors dropped off, pre- vs. post-redesign

Sustained usage over 30–90 days was unproven

Approach

Activation & Funnel Analysis · Feature Adoption Tracking · Retention Modelling

What Changed

Mapped the full activation funnel Traced 6 stages from access to submission to find exactly where users dropped off.

Defined activation as two valid paths Single-session and offline draft-to-sync completion both count as success.

Tracked feature adoption weekly Measured offline mode, auto-sync, and draft save adoption over an 8-week ramp.

Modeled 90-day retention Measured sustained submission behavior, not just first-time completion.

Reframed the core health metric Completion rate replaced logins and time-in-app as the primary signal.

Transformation Scale

6-stage funnel instrumented **3** core features tracked for adoption **8**-week adoption curve modeled

A measurable framework for whether field access actually became field usage.

Business Impact

4× activation rate vs. pre-redesign — 21% → 55% at final funnel stage

3.8× growth in offline-path completions — 5% → 38% of completions

3.8× higher 90-day retention — 17% → 64% sustained submission rate

Offline mode adoption reached 83% by week 8, up from 41% in week 1

Completion rate established as the core product health metric

Continue to Phase 3 — AI Enablement →

PHASE 3

Portfolio learning exercise — a product specification only, not built or delivered. Builds on Phase 1 (real) and Phase 2 (illustrative).

Field access was solved. Completion still depended on manual template search, typing, and self-checking.

[hero visual — agent architecture diagram]

15–20% targeted lift in field activation **20–30%** additional targeted reduction in time-to-value

Business Objective

Design an AI layer that helps assessors move faster and with more confidence at the point of inspection, without removing their control over what gets submitted.

Business Challenges

Template selection and long-form data entry still slowed assessors down

Manual capture kept transcription errors alive even after the redesign

Managers had no plain-language way to see where adoption was still breaking down

Approach

Agentic Workflow Design · Voice UX · Product Analytics · Guardrails & Governance

What Changed

Designed a smart template recommender Suggests the right template by GPS and history, replacing manual search.

Specified voice-to-form capture Assessors speak observations; the copilot maps them to structured fields.

Built a pre-submit validation agent Checks completeness before submission — assessor keeps final say.

Modeled adoption-risk scoring Flags assessors likely to abandon before they do, not after.

Specified a manager insights copilot Plain-language queries over field adoption data, no custom reports needed.

Transformation Scale

7 AI-assisted capabilities specified end-to-end **1** orchestrator coordinating **6** specialised agent tools **4**-phase rollout roadmap defined — Foundation → Capture → Intelligence → Admin

A specification for faster, more confident field capture, with a human always in the loop.

Business Impact

+15–20% targeted lift in field activation

Additional 20–30% targeted reduction in time-to-value beyond Phase 1

Abandonment rate targeted to hold at $\leq 7\%$, vs. 74% pre-redesign baseline

+10% targeted lift in sustained 30–90 day submission rate

Full audit trail and human-in-the-loop guardrails specified for every AI suggestion

← **Back to Phase 1**

New diagram assets (Phase 2 & Phase 3)

Four new SVG files were created to match Phase 1's visual weight — same file folder as this document:

- [phase2-activation-funnel.svg](#) — Desktop vs. iPad completion funnel, pre/post redesign
- [phase2-feature-adoption-curve.svg](#) — 8-week adoption curve for offline mode, auto-sync, draft save

- [phase2-retention-impact.svg](#) — before/after strip for time-to-submission, offline-path usage, 90-day retention
- [phase3-agent-architecture.svg](#) — Compliance Field Copilot orchestrator + 6-tool architecture

All four use the same color system (slate gray = before/baseline, teal = after/improved) so they read as one visual family with Phase 1's design artifacts. They're vector SVGs, so you can resize freely without quality loss, and edit colors/text directly if you want to match your site's palette.